AS-IS / TO-BE and Gap Analysis: RAG-Diary

> How people write and review notes today, the problems that creates, how RAG-Diary closes each gap, and the expected outcome, with every gap tied to a functional requirement in the SRS.

> Thai version: AS-IS-TO-BE.md

1. Document control

Item	Detail
Document	AS-IS / TO-BE and Gap Analysis
Version	1.0
Prepared	15 October 2026
Author	Business Analyst / System Analyst (project owner)
Status	Reviewed, round 1 (25 October 2026)
Related documents	PRODUCT-BRIEF, STAKEHOLDER-MATRIX, SRS, PROCESS-FLOWS

2. Purpose and scope

This document answers one question: before RAG-Diary, how did people write and review their notes, where did they get stuck, and what does the new system change? It shows where every requirement comes from and justifies the scope of version 1.0.

Scope: the writing and review process of a single user, across write, keep, retrieve and review, with the gap table and expected outcomes. Out of scope: formal BPMN process models (in PROCESS-FLOWS) and functional requirement detail (in the SRS).

3. AS-IS: the current process

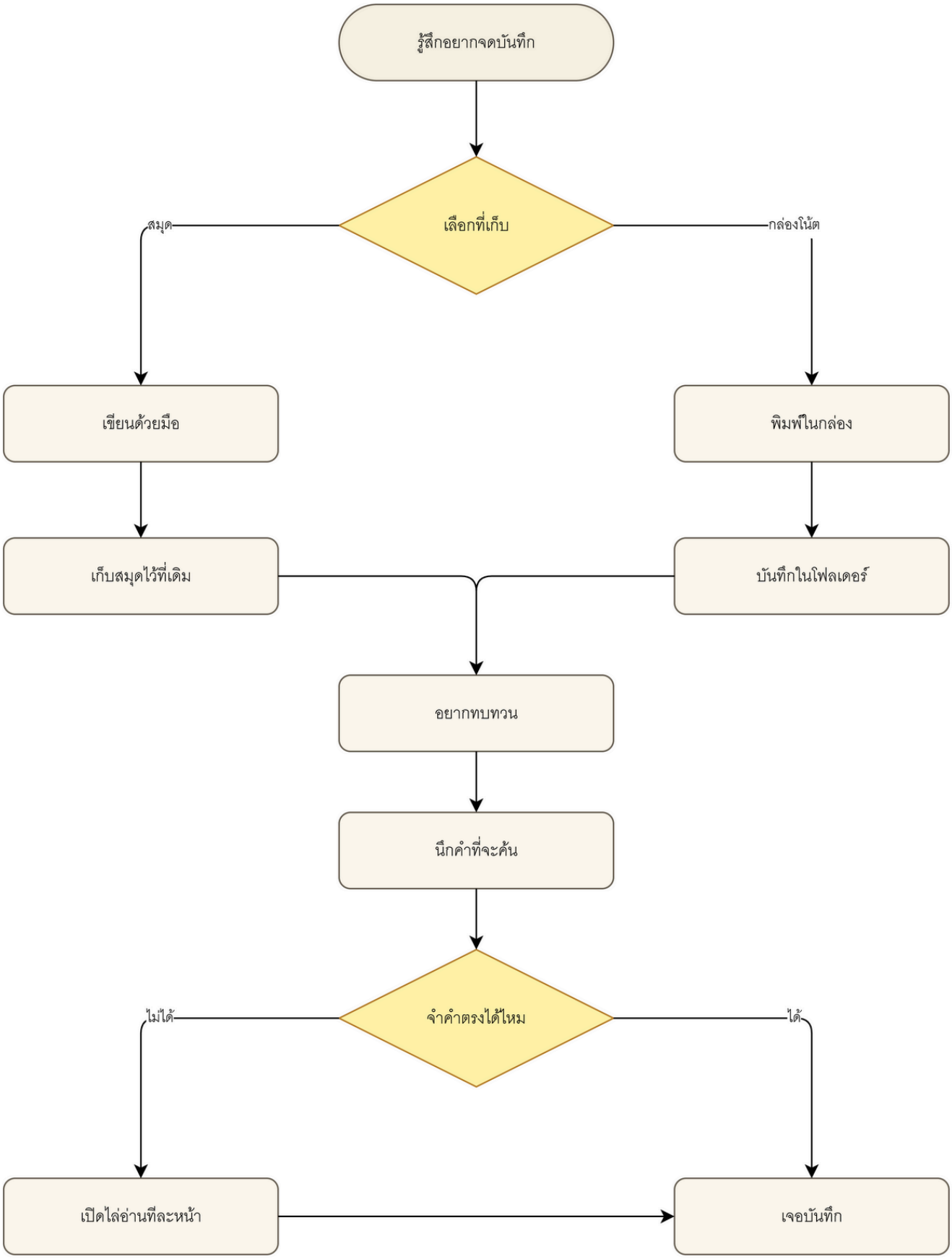
The current process is based on a general note app and a paper notebook, with no mechanism for retrieval by meaning.

3.1 Current steps

1. Write: by hand in a notebook, or typed into a note app on a phone.

2. Keep: scattered across several places, several notebooks and app folders, with inconsistent categorisation.
3. Retrieve: search by words that match the text; if the words are forgotten, flip through page by page.
4. Review: read back manually, with no overview and no trends.

3.2 AS-IS flow



3.3 Pain points

ID	Problem	Effect on the user	Supporting indicator (assumption)
PP-01	Keyword retrieval fails when the exact word is forgotten	The entry cannot be found	Keyword search succeeds about 40 percent of the time
PP-02	Searching takes a long time when reading through	Wastes time and reduces the wish to write	About 3 to 5 minutes per search
PP-03	Data is scattered across places	No single overview	Entries sit in 2 to 3 places per person
PP-04	Reviewing mood over time is hard	Trends in feeling are invisible	Everything must be read to summarise
PP-05	Sensitive data has no protection	Privacy is at risk	A paper notebook has no access control
PP-06	Important periods have no form to be kept in	Memories fade, with nothing to look back at	Only raw text survives, with no shape

> The figures in the indicator column are demonstration assumptions used to set targets. They do not come from field data collection.

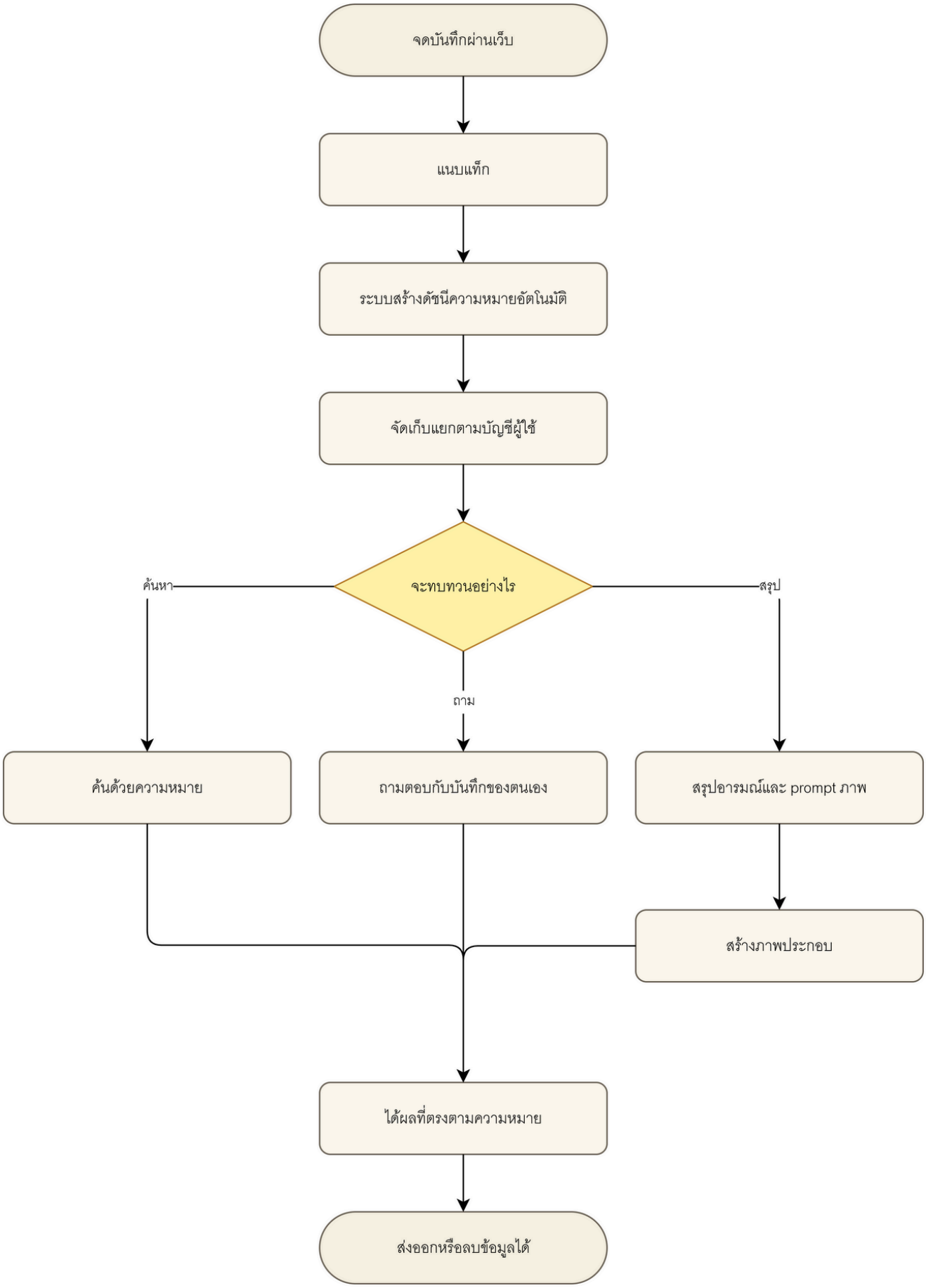
4. TO-BE: the target process

RAG-Diary brings writing, semantic storage and retrieval into one place, tied to a user account.

4.1 Target steps

1. Write: write an entry in the web app with a tag.
2. Keep: the system builds the semantic index automatically and stores it per account.
3. Retrieve: search by meaning, or ask a question in natural language about one's own entries.
4. Review: request a mood summary and an image prompt, then generate the image.
5. Manage data: export or delete everything at any time.

4.2 TO-BE flow



5. Gap analysis

ID	Gap from the as-is state	How the system closes it	Related requirements
GAP-01	Retrieval fails when the exact word is forgotten (PP-01)	Search by meaning instead of word matching	FR-05, FR-09
GAP-02	Searching takes a long time (PP-02)	Semantic search and natural-language question answering	FR-09, FR-10
GAP-03	Data is scattered (PP-03)	One place for all entries, with a list view	FR-06
GAP-04	Reviewing mood over time is hard (PP-04)	A mood summary as text, so trends become visible	FR-11
GAP-05	Sensitive data has no protection (PP-05)	Authentication, per-user isolation, and export or deletion	FR-01, FR-02, FR-03, FR-13, FR-14, FR-15
GAP-06	Writing and editing have no structure (a general observation, not from one pain point)	Write with a tag, and edit or delete per entry	FR-04, FR-07, FR-08
GAP-07	Important periods have no tangible place to be kept (PP-06)	A mood summary and a generated image that hold the period	FR-11, FR-12

6. Expected outcomes

Area	Before (AS-IS)	Target (TO-BE)
Retrieval	Relies on exact words	Found by meaning even when the words are forgotten
Time	Several minutes per search	Search and ask within seconds
Privacy	Scattered and uncontrolled	Isolated per user, deletable and exportable
Review	Read everything by hand	A mood summary and an illustration
Keeping memories	Raw text buried under newer entries	Important periods kept as a summary and an image

The numeric targets and how they are measured come from PRODUCT-BRIEF and the NFRs in the SRS.

7. Assumptions

- The figures for PP-01 and PP-02 are demonstration assumptions used to set design targets, not real data.

- The AS-IS comparison covers both paper notebooks and general note apps, so the picture includes users who have no semantic search at all.
- The process describes a single user. Sharing entries in future would change the privacy model.

8. References

Document	Relationship
PRODUCT-BRIEF	Business goals and measures
STAKEHOLDER-MATRIX	Stakeholder concerns, especially privacy
SRS	The functional requirements the gap analysis references
PROCESS-FLOWS	The formal process models at the next level